Allogenic Human Umbilical Cord Blood-Derived Mesenchymal Stem Cells Is More Effective Than Bone Marrow Aspiration Concentrate for Cartilage Regeneration After High Tibial Osteotomy in Medial Unicompartmental Osteoarthritis of Knee

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Purpose: The purpose of this study was to compare the outcome of cartilage regeneration between bone marrow aspirate concentrate (BMAC) augmentation and allogeneic human umbilical cord blood–derived mesenchymal stem cell (hUCB-MSCs) transplantation in high tibial osteotomy (HTO) with microfracture (MFX) for medial unicompartmental osteoarthritis (OA) of the knee in the young and active patient. **Methods:** Between January 2015 and December 2019, the patients who underwent HTO and arthroscopy with MFX combined with BMAC or allogeneic hUCB-MSCs procedure for medial unicompartmental OA with kissing lesion, which was shown full-thickness cartilage defect ($\geq$ International Cartilage Repair Society [ICRS] grade 3B) in medial femoral cartilage and medial tibial cartilage, were include in this study. Retrospectively we compared clinical outcomes, including Hospital for Special Surgery score, Knee Society Score (KSS) pain and function, and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) score between BMAC and hUCB-MSCs group at minimum of 1-year follow-up. Also, second-look arthroscopy was performed simultaneously with removal of the plate after complete bone union. Cartilage regeneration was graded by the ICRS grading system at second-look arthroscopy. Radiological measurement including hip-knee-ankle (HKA) angle, posterior tibial slope angle, and correction angle were assessed. **Results:** Of 150 cases that underwent HTO with MFX combined with BMAC or allogeneic hUCB-MSCs procedure for medial unicompartmental OA, 123 cases underwent plate removal and second-look arthroscopy after a minimum of 1 year after the HTO surgery. Seventy-four cases were kissing lesion in medial femoral cartilage and medial tibial cartilage during initial HTO surgery. Finally, the BMAC group composed of 42 cases and hUCB-MSCs group composed of 32 cases were retrospectively identified in patients who had kissing lesions and second-look arthroscopies with a minimum of 1 year of follow-up. At the final follow-up of mean 18.7 months (standard deviation = 4.6 months), clinical outcomes in both groups had improved. However, there were no significant differences between the IKDC, WOMAC, or KSS pain and function scores in the 2 groups ($P > .05$). At second-look arthroscopy, the ICRS grade was significantly better in the hUCB-MSC group than in the BMAC group in both medial femoral and medial tibial cartilage ($P = .001$ for both). The average ICRS grade of the BMAC group improved from 3.9 before surgery to 2.8 after surgery. The average ICRS grade of the hUBC-MSC group improved from 3.9 before surgery to 2.0 after surgery. Radiological findings comparing postoperative HKA angle, posterior tibial slope angle, and correction angle showed no significant differences between the groups ($P > .05$). Therefore it was found that the postoperative correction amount did not affect the postoperative cartilage regeneration results. **Conclusions:** We found that the hUCB-MSC procedure was more effective than the BMAC procedure for cartilage regeneration in medial unicompartmental knee OA even though the clinical outcomes improved regardless of which treatment was administered. **Level of Evidence:** Level III, retrospective comparative study.

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Abnormal distribution of weightbearing stress within the medial and lateral tibial plateau may lead to progressive cartilage degeneration and eventually unicompartmental knee osteoarthritis (OA).[1] Several procedures, such as high tibial osteotomy (HTO), unicompartmental knee arthroplasty, chondrocyte implantation, and mesenchymal stem cell (MSC) therapy, are available for middle-aged adults with severe pain and varus deformity of the knee joint who do not require total knee arthroplasty.[2] Therefore the ideal candidate for an HTO is a young patient (<60 years of age), with isolated medial OA, with good range of motion, and without ligamentous instability.[3] Furthermore, excessive stress from varus knee malalignment can worsen symptoms associated with OA or osteochondral lesions, which cause pain, disability, and eventually OA progression.[4] Treatments targeting articular cartilage defects include microfracture (MFX), osteochondral or chondrocyte transplantation, autologous bone marrow aspirate concentrate (BMAC), and MSC therapies.[3] The use of stem cells has become a major focal point in the field of regenerative medicine. Many human tissues, including bone marrow, adipose tissue, umbilical cord blood, and synovium, are well-known sources of adult MSCs.[5]

The combination of HTO and cartilage repair procedures has been broadly used for cartilage repair and alignment correction.[6] One such procedure, BMAC augmentation, is technically easy and fast and provides a safe and suitable source of autologous MSCs. Human umbilical cord blood–derived MSCs (hUCB-MSCs) are isolated in a noninvasive manner and are hypoimmunogenic.[7] Furthermore, hUCB-MSCs have shown high expansion capacity, providing enough cells for therapeutic applications.[8] Few studies have compared clinical outcomes and cartilage regeneration using BMAC and hUCB-MSCs after HTO.[9]

The purpose of this study was to compare the outcome of cartilage regeneration between BMAC augmentation and hUCB-MSCs transplantation in HTO with MFX for medial unicompartmental OA of knee in the young and active patient. Therefore a retrospective, single-blind study was designed to analyze the outcome of cartilage regeneration between 2 procedures. We hypothesized that HTO with hUCB-MSC procedures will show superior results in cartilage regeneration and clinical outcomes than HTO with BMAC augmentation.

Methods

Patient Selection

This retrospective study was conducted at 1 hospital between January 2015 and December 2019 after obtaining Institutional Review Board and Ethics Committee approval. We included the patients who underwent HTO with MFX combined with BMAC or allogeneic hUCB-MSCs procedure because of severe medial-side knee pain and varus deformity of lower leg. Among them, we included the patients who underwent plate removal and second-look arthroscopy to check cartilage regeneration after the complete bony union at least at the 1-year follow-up period. We enrolled patients with kissing lesions showing full-thickness cartilage defects (≥International Cartilage Repair Society (ICRS) grade 3B) in the medial femoral and medial tibial cartilage at initial surgery.[10]

We excluded patients with severe obesity (body mass index [BMI] ≥ 35),[11] as well as those who had previous ipsilateral knee fractures, other previous cartilage repair procedures (such as chondroplasty, drilling, or stem cell transplantation other than BMAC), HTO combined with anterior cruciate ligament reconstruction or partial lateral meniscectomy, presence of ICRS grade 2+ cartilage lesions affecting the lateral or patellofemoral compartments, or less than 1 year's follow-up duration.

HTO for medial compartment OA were performed at our institution by 2 experienced orthopaedic surgeons (E.K.S., J.K.S.) using the same techniques. At the time of preoperative examination, the patient was given a sufficient explanation of both methods, and the patient was allowed to choose between the 2 cartilage regeneration methods.

In the BMAC group, HTO with MFX combined with BMAC augmentation was carried out. In the hUCB-MSC group, HTO with MFX combined with hUCB-MSC transplantation was carried out.

Operative Technique

The surgery was performed at 1 institution by 2 experienced orthopaedic surgeons (E.K.S., J.K.S.) using the same techniques. In the BMAC group, bone marrow aspiration was performed with the patient under spinal anesthesia at the contralateral anterior superior iliac spine. A 60 mL heparinized syringe was used, and at least 40 mL of bone marrow was required. Bone marrow was transferred to a 50 mL sterilized tube that would be spun in a centrifuge for 4 minutes at 2,500 rpm for marrow cell concentration. The concentrated bone marrow, which theoretically contained a relatively high proportion of MSCs, was collected with another 60 mL heparinized syringe.[12]

The hUCB-MSC group was treated with a medicinal product (Cartistem, hUCB-MSCs-HA hydrogel composites) provided as a new investigational drug by Medipost and produced under good manufacturing practice guidelines. Therapeutic use of Cartistem for cartilage repair was reviewed and approved by the Korea Food and Drug Administration in January 2012.[13] Human umbilical cord blood was collected from umbilical veins at the time of neonatal delivery with informed consent from the mothers and stored in

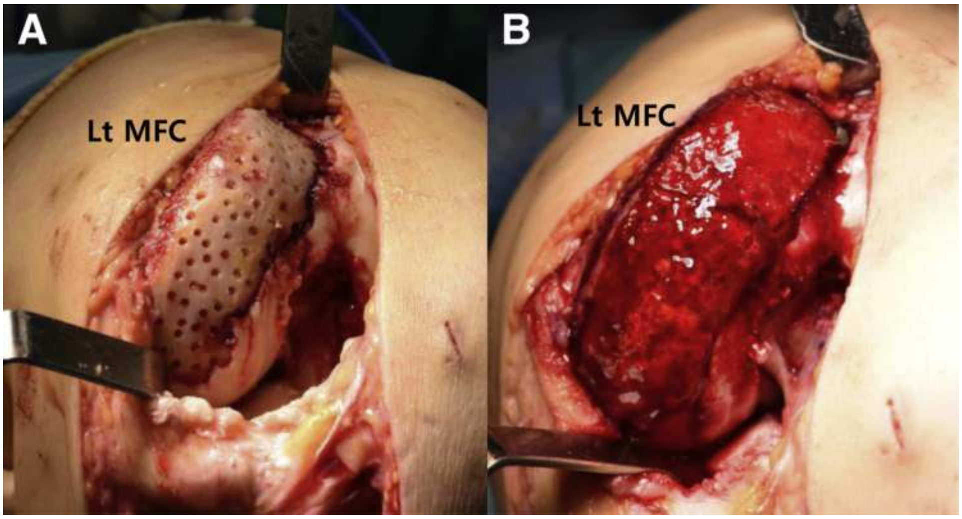

Fig 1. Bone marrow aspirate concentrate (BMAC) implantation in left medial femoral condyle (Lt MFC). (A) Prepared bed of lesion (microfracture with multiple drilling). (B) Fibrin sealant patch soaked with the prepared BMAC was fixed with fibrin glue.

a cord blood bank. The hUCB-MSCs were isolated and characterized according to previously published methods.[14]

After the patients were in a supine position and a tourniquet was applied, diagnostic arthroscopy was performed. During arthroscopy, debridement and evaluation of cartilage defects were carried out with a shaver and small-size ring curette to obtain a definitive, intact cartilaginous rim. Also, arthroscopic results were collected by the senior 2 authors. Biplanar opening wedge osteotomy was then performed.[15] Knee alignment was checked before surgery using teleroentgenography while patients were in a standing position to evaluate the correction angle and osteotomy size. The Fujisawa point was used to determine the weightbearing line to correct mechanical alignment using intraoperative fluoroscopy.[16] A cancellous allograft filled the open wedge only in some cases, as determined during surgery by the surgeons. After HTO, mini-arthrotomy exposed the medial femoral condyle. In the BMAC group, after MFX with multiple drilling (2 mm diameter and 5 mm depth) was performed, a fibrin sealant patch (TachoSil; Takeda Pharma A/S) soaked with the prepared BMAC was fixed with fibrin glue (Greenplast kit; Green Cross) (Fig 1).[17] In the hUCB-MSC group, multiple drill holes (4 and 2 mm in diameter and 5 mm depth) were made approximately 2 to 3 mm apart at the cartilage defect site of the femoral condyle. The drug was implanted into the holes in the lesion from the base to the surface (Fig 2).[18] In the case of kissing lesions, the composite was only transplanted into femoral lesions because of dose limitations in terms of cell numbers.

Rehabilitation started on postoperative day 1. The use of a knee brace facilitated exercise of the quadriceps and ankle movement. Range of motion of the knee joint was

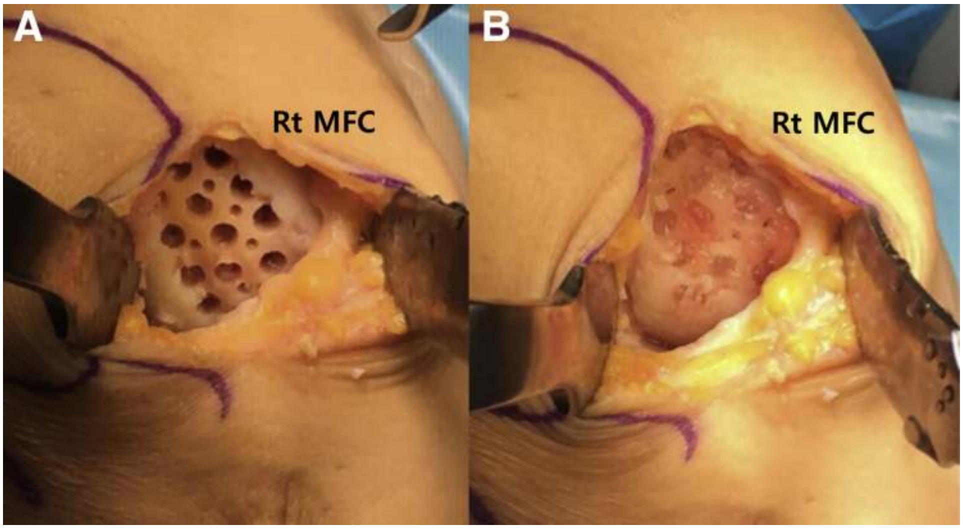

Fig 2. Human umbilical cord blood−derived mesenchymal stem cell (hUCB-MSC) implantation in right medial femoral condyle (Rt MFC). (A) Prepared bed of lesion (microfracture with multiple drilling). (B) The hUCB-MSC implanted into the holes in the lesion from the base to the surface.

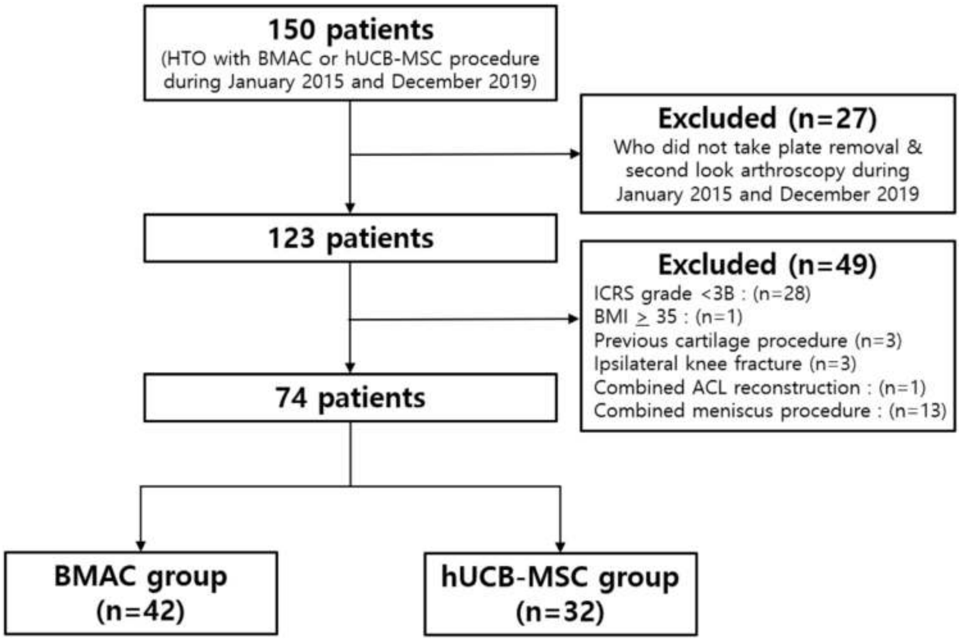

Fig 3. Patient selection flowchart. ACL, anterior cruciate ligament; BMAC, bone marrow aspirate concentrate; BMI, body mass index; hUCB-MSC, human umbilical cord blood–derived mesenchymal stem cell; ICRS, International Cartilage Repair Society.

limited to 0° to 30°. Gradually, knee flexion from 0° to 90° was allowed after 2 postoperative weeks, whereas non-weightbearing to partial weightbearing walking with crutches gait of isometric exercises was advised after 6 weeks. A total of 12 weeks of non-weightbearing to partial-weightbearing periods, full weightbearing was advised depending on the level of bone healing according to radiographic findings after 12 weeks.

After confirmation of union at the osteotomy sites by anterior-posterior radiographs, second-look arthroscopy was performed simultaneously with removal of plates after bone union. Arthroscopic results were collected again by the 2 senior authors.

Evaluation of Clinical Outcomes, Radiologic Outcomes, and Cartilage Regeneration

All patients were clinically evaluated using the following indices preoperatively and at the final follow-up: Hospital for Special Surgery (HSS) score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) score, and Knee Society Score (KSS) for pain and function. Because we thought that if the radiographic results were different in the 2 groups, it would affect the outcomes of cartilage regeneration after surgery, we checked the radiologic parameters before and after surgery. The hip-knee-ankle (HKA) angle was measured by teleroentgenography, and the posterior tibial slope (PTS) was measured by a lateral plain radiograph before the operation and at latest follow-up. For HKA angles in varus alignment, values were marked negatively. In addition, correction angles were recorded before surgery for demographic analysis. Knee cartilage lesions were assessed by arthroscopy during surgery and their recovery after surgery according to the ICRS grading system.[10]

Table 1. Comparison of Preoperative Demographics

Parameters	BMAC Group (n = 42)	hUCB-MSC Group (n = 32)	*P* value
Sex* (M/F)	6/36	6/26	.53
Age† (yr)	60.7 ± 4.1	58.1 ± 3.6	.19
Body mass index† (kg/m^2)	26.1 ± 2.8	26.6 ± 3.0	.122
Follow-up period† (months)	20.7 ± 6.1	15.6 ± 2.8	.08
Defect size of cartilage† (MFC, cm^2)	6.5 ± 2.9	7.0 ± 1.9	.18
Mean ICRS grade			
MFC	3.9 ± 0.3	3.9 ± 0.3	.34
MTC	3.9 ± 0.3	3.9 ± 0.3	.77

Data are presented median ± standard deviation. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.

BMAC, bone marrow aspirate concentrate; hUCB-MSC, human umbilical cord blood–derived mesenchymal stem cell; ICRS, International Cartilage Repair Society; MFC, medial femoral condyle; MTC, medial tibial condyle.

*The Pearson's χ^2 test.

†The Independent t-test.

Table 2. Proportions of Patients Achieving MCID for Clinical Outcomes

Parameters	MCID	BMAC Group (n = 42)	hUCB-MSC Group (n = 32)	*P* Value
HSS	8.29	24 (57.14%)	21 (65.63%)	.09
WOMAC	15.0	23 (54.76%)	21 (65.63%)	.51
KSS pain	3.0	23 (54.76%)	24 (75%)	.48
KSS function	5.6	25 (59.52%)	22 (68.75%)	.41

The Pearson's χ^2 square test. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.

BMAC, bone marrow aspirate concentrate; hUCB-MSC, human umbilical cord blood−derived mesenchymal stem cell; HSS, Hospital for Special Surgery; KSS, Knee Society Score; MCID, mean clinically important difference; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

Two orthopaedic surgeons (J.K.K., H.W.A.) who did not take part in the surgeries and were blinded to the experimental design performed all measurements, including clinical scores, radiological measurements, and ICRS grade according to the arthroscopic results.

Statistical Analysis

Analysis was performed using IBM SPSS Statistics for Windows (version 22; IBM). For analysis of demographic data, independent *t*-tests were used to compare age, BMI, follow-up period, cartilage defect size, preoperative HKA angle, and PTS between the 2 groups. The χ^2 test was used to analyze categorical variables. Comparison of preoperative and postoperative clinical outcomes was performed using the Mann-Whitney U test. Additionally, minimal clinically important difference (MCID) was checked. It is clinically important to estimate whether the clinical outcome score improved beyond a MCID, the change in patients who identified themselves as minimally better or minimally worse as a result of the surgical procedure. Fisher's exact test was used to analyze ICRS grade at second-look arthroscopy. We defined *P* values <0.05 as statistically significant.

Results

Demographics

During the study period, 150 patients who underwent HTO with MFX combined with BMAC or allogeneic hUCB-MSCs procedure performed by 2 surgeons were considered for inclusion. Of 150 cases, 123 cases underwent plate removal and second-look arthroscopy to check cartilage regeneration after a minimum of 1 year after initial HTO surgery.

In total, 74 cases were kissing lesions showing full-thickness cartilage defects (≥ International Cartilage Repair Society [ICRS] grade 3B) in medial femoral cartilage and medial tibial cartilage during initial HTO surgery. In the BMAC group, HTO with MFX combined with BMAC augmentation was carried out in 42 patients. In the hUCB-MSC group, HTO with MFX combined with hUCB-MSC transplantation was carried out in 32 patients (Fig 3).

As shown in Table 1, patient demographic data were similar in both groups. There were no significant differences in age, sex, BMI, clinical outcomes, or radiological measurements between the 2 groups (Table 1).

Table 3. Comparison of Clinical Outcomes

Parameters	BMAC Group (n = 42)	hUCB-MSC Group (n = 32)	*P* Value
HSS			
Preoperative	57.9 ± 12.9	56.1 ± 10.6	.826
Postoperative	79.2 ± 11.5	84.6 ± 15.5	.041
WOMAC			
Preoperative	43.9 ± 12.7	45.2 ± 8.8	.697
Postoperative	23.4 ± 11.6	19.5 ± 15.8	.080
KSS pain			
Preoperative	30.8 ± 11.0	31.6 ± 10.4	.993
Postoperative	40.6 ± 9.1	42.8 ± 7.9	.380
KSS function			
Preoperative	62.3 ± 11.9	63.1 ± 11.2	.766
Postoperative	80.1 ± 15.0	82.4 ± 15.5	.437

Mann-Whitney U test. Data are presented median ± standard deviation. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.

BMAC, bone marrow aspirate concentrate; hUCB-MSC, human umbilical cord blood−derived mesenchymal stem cell; HSS, Hospital for Special Surgery; KSS, Knee Society Score; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

Table 4. Comparison of Radiological Outcomes

Parameters	BMAC Group (n = 42)	hUCB-MSC Group (n = 32)	*P* Value
HKA angle			
Preoperative (varus, °)	8.6 ± 3.1	7.4 ± 2.6	.194
Postoperative (valgus, °)	2.8 ± 3.2	2.9 ± 1.6	.122
Posterior tibial slope			
Preoperative (°)	8.5 ± 3.9	7.6 ± 3.7	.078
Postoperative (°)	8.8 ± 4.5	7.4 ± 3.8	.183

Independent *t*-test. Data are presented median ± standard deviation. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.

BMAC, bone marrow aspirate concentrate; HKA angle, hip-knee-ankle angle; hUCB-MSC, human umbilical cord blood–derived mesenchymal stem cell.

Clinical and Radiologic Outcomes

According to the previous studies, the MCIDs of the HSS, WOMAC, KSS pain and function scores were 8.29, 15.0, 3.0 and 5.6 points.[19-21] In this study, the mean improvement of the total HSS, WOMAC, KSS pain, and function scores were 21.3, 20.5, 9.8, and 17.8 points in the BMAC group and 28.5, 25.7, 11.2, and 19.3 points in the hUCB-MSC group. The proportions of patients achieving the MCID for each clinical outcome are illustrated in Table 2.

During the follow-up periods (BMAC, 20.7 ± 6.1 months; hUCB-MSC, 15.6 ± 2.8 months), clinical outcomes, including the HSS score, WOMAC score, and KSS pain and function score, improved in both groups after surgery. However, there was no statistically significant difference between postoperative clinical outcomes in the 2 groups (Table 3).

As shown in Table 4, there were no statistically significant differences between the groups (P = .122, 0.183) in terms of the postoperative HKA angle (BMAC, 2.8 ± 3.2; hUBC-MSC, 2.9 ± 1.6) or PTS (BMAC, 8.8 ± 4.5; hUBC-MSC, 7.4 ± 3.8). Therefore it was found that the postoperative correction amount of varus alignment did not affect the postoperative cartilage regeneration results.

Analysis of Arthroscopic Findings

At least 1 year after initial surgery, 123 patients (77 in the BMAC group (Fig 4) and 46 in the hUCB-MSC group (Fig 5)) underwent second-look arthroscopy along with plate removal. Among them, 74 had kissing lesions showing full-thickness cartilage defects (with ICRS grades >3B) in the medial femoral and medial tibial cartilage at initial surgery.

Second-look arthroscopic findings related to articular cartilage regeneration according to the ICRS grade are summarized in Tables 5 and 6. The hUCB-MSC group showed better cartilage regeneration than did the BMAC group at both the medial femoral and medial tibial condyles.

In addition, we compared the differences in clinical results according to the degree of cartilage regeneration observed during second-look arthroscopy. Groups of patients with ICRS grade 1-2 and ICRS grade 3-4 regeneration were compared. There were no statistically significant differences in KSS pain and function

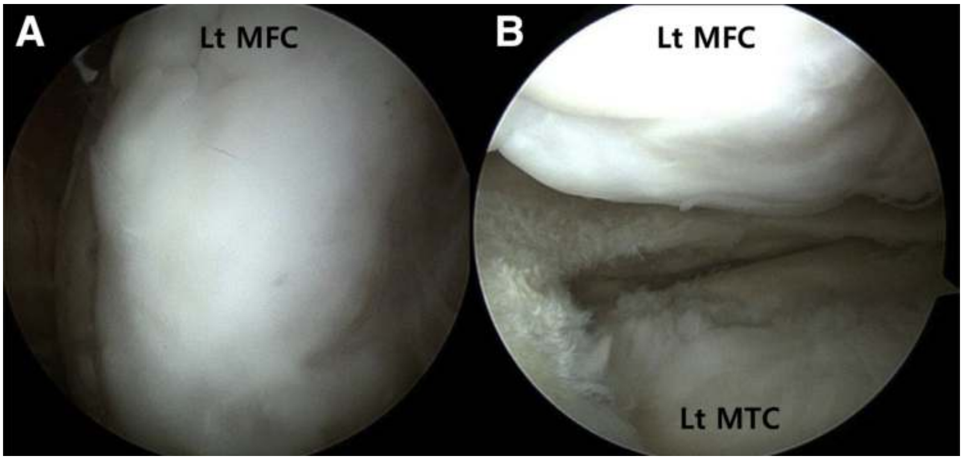

Fig 4. Findings from second look arthroscopy at 1 year after bone marrow aspirate concentrate implantation in left medial femoral condyle (Lt MFC). (A) The defect site of MFC was resurfaced well with regenerated cartilage showing a smooth surface. Good integration with the surrounding cartilage was also observed. The repair cartilage was assessed as grade 1 according to the International Cartilage Repair Society (ICRS) grade. (B) The defect site of left medial tibial condyle (Lt MTC) was also resurfaced with regenerated cartilage showing a fraying surface. The repair cartilage was assessed as grade 2 according to the ICRS grade.

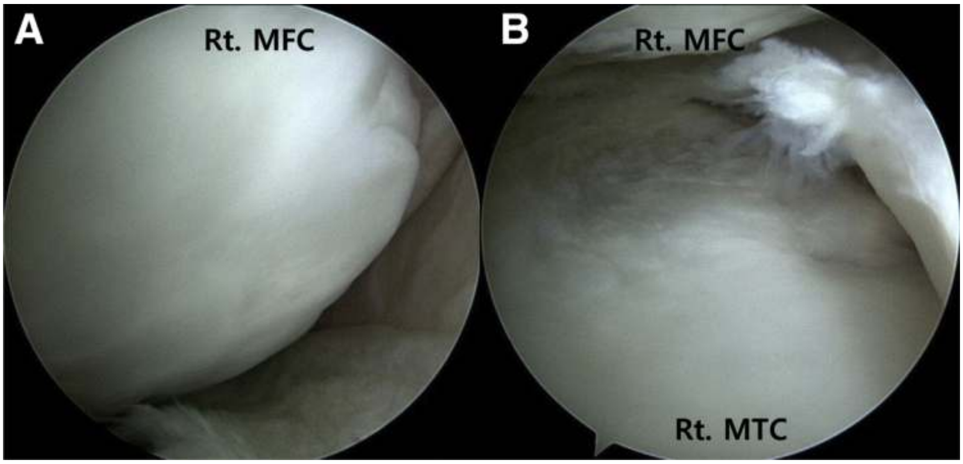

Fig 5. Findings from second-look arthroscopy at 1 year after human umbilical cord blood−derived mesenchymal stem cell implantation in right medial femoral condyle (Rt. MFC). (A) The defect site of MFC was resurfaced well with regenerated cartilage showing a smooth surface. Good integration with the surrounding cartilage was also observed. The repair cartilage was assessed as grade 1 according to the International Cartilage Repair Society (ICRS) grade. (B) The defect site of medial tibial condyle (MTC) was also resurfaced, with regenerated cartilage showing a fraying surface. The repair cartilage was assessed as grade 2 according to the ICRS grade.

scores, but the ICRS grade 1-2 group showed better clinical results in terms of WOMAC and HSS scores (Table 7).

Discussion

The main finding of this study was that the clinical outcomes improved regardless of which treatment was administered. However, the hUCB-MSC procedure was more effective than the BMAC augmentation for cartilage regeneration in medial unicompartmental knee OA.

In our study, clinical outcomes after surgery were significantly improved in both groups compared to baseline measurements before surgery. Medial knee joint pain improved because HTO reduced the load on the medial side of the knee joint by moving the joint reaction force from the medial to the lateral side of the knee. In addition, anti-inflammatory effects caused by cartilage regeneration via the BMAC or hUCB-MSC procedures and pain relief caused by paracrine effects should be considered.

ICRS grades were significantly higher in the hUCB-MSC group than in the BMAC group at second-look arthroscopy. We think that the hUCB-MSC procedure had a more potent effect than the BMAC procedure on cartilage regeneration because of its higher proliferation rate, karyotype stability after expansion, and chondrogenic potential.

Although hUCB-MSCs produced better cartilage regeneration effects according to the ICRS grade, there were no significant differences between clinical results in the groups over the short-term follow-up period (mean 18.7 months). In addition, because hUCB-MSC transplantation costs approximately $5,300 US more than does BMAC on average, the choice of surgical treatment should be left to the patient.

Recently, extraction and subsequent concentration of MSCs obtained from autologous bone marrow was introduced, and this represents the next generation in injectable intra-articular orthobiological therapies for patients with cartilage disease.[22] BMAC has emerged as an important biological tool for orthopaedic surgeons because it is one of the few forms of stem cell and growth factor delivery currently approved by the United States Food and Drug Administration.[23] The procedure is technically easy and fast, provides a safe and suitable source of MSCs, and enables harvest and intraoperative transplantation in one sitting.[24] BMAC is obtained through density gradient centrifugation of bone marrow aspirate typically aspirated from the iliac

Table 5. Comparison of ICRS Grade at Second-Look Arthroscopy in the Medial Femoral Condyle

Grade	BMAC group (n = 42)	hUCB-MSC group (n = 32)	*P* Value
Grade I	1 (2.4%)	6 (18.8%)	.001
Grade II	18 (42.6%)	20 (62.4%)	.001
Grade III	12 (28.6%)	6 (18.8%)	.001
Grade IV	11 (26.4%)	0	.001

Fisher exact test. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.
BMAC, bone marrow aspirate concentrate; hUCB-MSC, human umbilical cord blood−derived mesenchymal stem cell.

Table 6. Comparison of ICRS Grade at Second-Look Arthroscopy in the Medial Tibial Condyle

Grade	BMAC group (n = 42)	hUCB-MSC group (n = 32)	*P* Value
Grade I	1 (2.4%)	2 (6.3%)	.001
Grade II	16 (38.1%)	24 (75%)	.001
Grade III	9 (21.4%)	5 (15.6%)	.001
Grade IV	16 (38.1%)	1 (3.1%)	.001

Fisher exact test. The *P* values are of intergroup comparisons, with $P < .05$ indicating significance.
BMAC, bone marrow aspirate concentrate; hUCB-MSC, human umbilical cord blood–derived mesenchymal stem cell.

crest.[25] Because only about 0.001% of nucleated cells from BMA are MSCs,[26] attempts are made to increase their number, usually by concentrating the autologous BMA by density-gradient centrifugation.[27] BMAC provides elevated levels of hematopoietic stem cells, MSCs, platelets, chemokines, and cytokines, including platelet-derived growth factor and transforming growth factor-β.[28] These growth factors are not only contained within the alpha granules of platelets but also secreted by MSCs and can induce their chondrogenesis.[29] The growth factors also initiate stem cell migration to injury sites and provide adhesion sites for the migrating stem cells.[30] Moreover, BMAC possesses anti-inflammatory and immunomodulatory properties that have potential anabolic and anti-inflammatory effects, enhancing cartilage repair.[31] A previous study demonstrated the good efficacy of BMAC in treating focal cartilage defects, most of which were large cartilage lesions (>3 cm^2).[23] Over a short-term follow-up period, better BMAC outcomes have been correlated with younger age (<45 years), smaller chondral lesion size, and fewer lesions, with good coverage of defects as observed by magnetic resonance imaging or second-look arthroscopy.[32]

But, in BMAC, mesenchymal cells are present but lack stem cell differentiation activities, thus not meeting the criteria for characterization as MSCs[33]; also, no standard technique or protocol for BMA harvesting exists. The commercially available systems including centrifuges do not provide equivalent cell numbers and concentrations. The ideal volume of required BMAC for the treatment of a specific defect volume remains to be determined. Standardizations of all of these factors are needed, because they may provide a more scientific way of evaluating possible effects, especially because there is a large interindividual variety in the number of produced and secreted growth factors by the many cells within the BMA.[27] Overall, although BMAC may have some regenerative potential for cartilage, it is not enough to fully restore hyaline cartilage.[33]

Although there are slight differences between studies, hUCB-MSCs have various advantages in that they are relatively easy to collect and have a similar expansion capacity to that of bone marrow– or adipose-derived MSCs.[7,8] The hUCB-MSCs have many more advantages as a cell source in cellular therapy. First, they have shown more than 1,000-fold expansion capacity as compared to that of bone marrow–derived MSCS.[14] Second, they cause less donor site morbidity than do bone marrow–derived MSCs, which are commonly harvested from the posterior iliac bone. Finally, hUCB-MSCs are less immunogenic and can escape host immune surveillance, provoking no obvious signs of immunological response even after xenotransplantation.[34]

The hUCB-MSCs have a higher proliferation rate, karyotype stability after prolonged expansion, and chondrogenic potential than BM-MSCs.[35] Moreover, hUCB-MSCs have several other advantages, including ease of availability from cord blood banks, noninvasive collection procedures, and hypoimmunogenic properties.[36] Some preclinical studies have demonstrated the efficacy of hUCB-MSCs in cartilage regeneration.[37] Thus we believe that hUCB-MSCs are strong candidates for the development of a novel stem cell–based medicinal product for cartilage regeneration in osteoarthritic patients.

Limitation

Certain limitations of this study should be considered. First, a relatively small number of patients participated in this clinical trial. Small sample sizes may have less power to detect significant effects, because patients' data for each group were collected under the strict

Table 7. Comparison of postoperative clinical outcomes according to the cartilage regeneration.

Parameters	Grade I&II group (n=45)	Grade III&IV group (n=30)	P-value
HSS	85.65 ± 8.17	75.29 ± 14.56	0.001
WOMAC	18.47 ± 12.52	26.86 ± 14.26	0.001
KSS pain	42.35 ± 7.41	40.48 ± 10.35	0.471
KSS function	83.24 ± 12.24	77.62 ± 18.88	0.315

Mann-Whitney U test. Data are presented median ± standard deviation. The p-values are of inter-group comparisons, with $p < 0.05$ indicating significance inclusion criteria as a retrospective study. Second, as with any retrospective study, there was no process for randomization in patient selection and treatment methods. This study can potentially be affected by selection and performance bias because of the lack of randomization. Third, although the sampling and injection processes were carried out consistently, no laboratory analysis measured cell numbers, growth factors, or other constituents in the aspirated concentrates. Fourth, this study evaluated the state of cartilage regeneration on the basis of clinical indexes an average of 18.7 month after surgery and thus only reported the results of a short-term follow-up period. Hence, future studies with mid- to long-term follow-up periods over 5 years will be needed.

Conclusion

We found that the hUCB-MSC procedure was more effective than the BMAC procedure for cartilage regeneration in medial unicompartmental knee OA even though the clinical outcomes improved regardless of which treatment was administered.

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Reference

1. Kim YS, Koh YG. Comparative matched-pair analysis of open-wedge high tibial osteotomy with versus without an injection of adipose-derived mesenchymal stem cells for varus knee osteoarthritis: Clinical and second-look arthroscopic results. *Am J Sports Med* 2018;46:2669-2677.
2. Jayaram P, Ikpeama U, Rothenberg JB, Malanga GA. Bone marrow-derived and adipose-derived mesenchymal stem cell therapy in primary knee osteoarthritis: A narrative review. *PM R* 2019;11:177-191.
3. Lee DC, Byun SJ. High tibial osteotomy. *Knee Surg Relat Res* 2012;24:61-69.
4. Biant LC, McNicholas MJ, Sprowson AP, Spalding T. The surgical management of symptomatic articular cartilage defects of the knee: Consensus statements from United Kingdom knee surgeons. *Knee* 2015;22:446-449.
5. Tuan RS, Boland G, Tuli R. Adult mesenchymal stem cells and cell-based tissue engineering. *Arthritis Res Ther* 2003;5:32-45.
6. Kim MS, Koh IJ, Choi YJ, Pak KH, In Y. Collagen augmentation improves the quality of cartilage repair after microfracture in patients undergoing high tibial osteotomy: A randomized controlled trial. *Am J Sports Med* 2017;45:1845-1855.
7. Flynn A, Barry F, O'Brien T. UC blood-derived mesenchymal stromal cells: An overview. *Cytotherapy* 2007;9: 717-726.
8. Kern S, Eichler H, Stoeve J, Klüter H, Bieback K. Comparative analysis of mesenchymal stem cells from bone marrow, umbilical cord blood, or adipose tissue. *Stem Cells* 2006;24:1294-1301.
9. Huh SW, Shetty AA, Ahmed S, Lee DH, Kim SJ. Autologous bone-marrow mesenchymal cell induced chondrogenesis (MCIC). *J Clin Orthop Trauma* 2016;7:153-156.
10. Brittberg M, Winalski CS. Evaluation of cartilage injuries and repair. *J Bone Joint Surg Am* 2003;85-A:58-69 (Suppl 2).
11. WHO. Global database on body mass index: An interactive surveillance tool for monitoring nutrition transition. *Public Health Nutr* 2006;9:658-660.
12. Cotter EJ, Wang KC, Yanke AB, Chubinskaya S. Bone marrow aspirate concentrate for cartilage defects of the knee: From bench to bedside evidence. *Cartilage* 2018;9: 161-170.
13. Park YB, Ha CW, Lee CH, Yoon YC, Park YG. Cartilage regeneration in osteoarthritic patients by a composite of allogeneic umbilical cord blood-derived mesenchymal stem cells and hyaluronate hydrogel: Results from a clinical trial for safety and proof-of-concept with 7 years of extended follow-up. *Stem Cells Transl Med* 2017;6: 613-621.
14. Yang SE, Ha CW, Jung M, et al. Mesenchymal stem/progenitor cells developed in cultures from UC blood. *Cytotherapy* 2004;6:476-486.
15. Floerkemeier S, Staubli AE, Schroeter S, Goldhahn S, Lobenhoffer P. Outcome after high tibial open-wedge osteotomy: A retrospective evaluation of 533 patients. *Knee Surg Sports Traumatol Arthrosc* 2013;21:170-180.
16. Fujisawa Y, Masuhara K, Shiomi S. The effect of high tibial osteotomy on osteoarthritis of the knee. An arthroscopic study of 54 knee joints. *Orthop Clin North Am* 1979;10:585-608.
17. Enea D, Cecconi S, Calcagno S, et al. Single-stage cartilage repair in the knee with microfracture covered with a resorbable polymer-based matrix and autologous bone marrow concentrate. *Knee* 2013;20:562-569.
18. Song JS, Hong KT, Kim NM, et al. Implantation of allogenic umbilical cord blood-derived mesenchymal stem cells improves knee osteoarthritis outcomes: Two-year follow-up. *Regen Ther* 2020;14:32-39.
19. Singh JA, Schleck C, Harmsen WS, Lewallen D. Validation of the Hospital for Special Surgery Knee questionnaire: Convergent validity, responsiveness and sensitivity to change. 2013 ACR/ARHP Annual Meeting
20. Escobar A, Quintana JM, Bilbao A, Aróstegui I, Lafuente I, Vidaurreta I. Responsiveness and clinically important differences for the WOMAC and SF-36 after total knee replacement. *Osteoarthritis Cartilage* 2007;15: 273-280.
21. Jacquet C, Pioger C, Khakha R, et al. Evaluation of the "Minimal Clinically Important Difference" (MCID) of the KOOS, KSS and SF-12 scores after open-wedge high tibial osteotomy [published online April 27, 2020]. *Knee Surg Sports Traumatol Arthrosc*. https://doi.org/10.1007/s00167-020-06026-0.
22. Szychlinska MA, Stoddart MJ, D'Amora U, Ambrosio L, Alini M, Musumeci G. Mesenchymal stem cell-based cartilage regeneration approach and cell senescence: Can we manipulate cell aging and function? *Tissue Eng Part B Rev* 2017;23:529-539.

23. Chahla J, Dean CS, Moatshe G, Pascual-Garrido C, Cruz RS, LaPrade RF. Concentrated bone marrow aspirate for the treatment of chondral injuries and osteoarthritis of the knee: A systematic review of outcomes. *Orthop J Sports Med* 2016;4:2325967115625481.
24. Turner LG. Federal regulatory oversight of US clinics marketing adipose-derived autologous stem cell interventions: Insights from 3 new FDA draft guidance documents. *Mayo Clin Proc* 2015;90:567-571.
25. Jäger M, Hernigou P, Zilkens C, et al. Cell therapy in bone healing disorders. *Orthop Rev(Pavia)* 2010;2:e20.
26. Kasten P, Beyen I, Egermann M, et al. Instant stem cell therapy: Characterization and concentration of human mesenchymal stem cells in vitro. *Eur Cell Master* 2008;16:47-55.
27. Madry H, Gao L, Eichler H, Orth P, Cucchiarini M. Bone marrow aspirate concentrate-enhanced marrow stimulation of chondral defects [published online May 14, 2017]. *Stem Cells Int*. https://doi.org/10.1155/2017/1609685.
28. McCarrel T, Fortier L. Temporal growth factor release from platelet-rich plasma, Trehalose lyophilized platelets, and bone marrow aspirate and their effect on tendon and ligament gene expression. *J Orthop Res* 2009;27:1033-1042.
29. Chen FH, Tuan RS. Mesenchymal stem cells in arthritic diseases. *Arthritis Res Ther* 2008;10:223.
30. Steinert AF, Rackwitz L, Gilbert F, Nöth U, Tuan RS. Concise review: the clinical application of mesenchymal stem cells for musculoskeletal regeneration: Current status and perspectives. *Stem Cells Transl Med* 2012;1:237-247.
31. Sampson S, Botto-van Bemden A, Aufiero D. Autologous bone marrow concentrate: Review and application of a novel intra-articular orthobiologic for cartilage disease. *Phys Sportsmed* 2013;41:7-18.
32. Gobbi A, Karnatzikos G, Sankineani SR. One-step surgery with multipotent stem cells for the treatment of large full-thickness chondral defects of the knee. *Am J Sports Med* 2014;42:648-657.
33. Cavinatto L, Hinckel BB, Tomlinson RE, Gupta S, Farr J, Bartolozzi AR. The role of bone marrow aspirate concentrate for the treatment of focal chondral lesions of the knee: A systematic review and critical analysis of animal and clinical studies. *Arthroscopy* 2018;35:1860-1877.
34. Ding Y, Bushell A, Wood KJ. Mesenchymal stem-cell immunosuppressive capabilities: Therapeutic implications in islet transplantation. *Transplantation* 2010;89:270-273.
35. Zhang X, Hirai M, Cantero S, et al. Isolation and characterization of mesenchymal stem cells from human umbilical cord blood: Reevaluation of critical factors for successful isolation and high ability to proliferate and differentiate to chondrocytes as compared to mesenchymal stem cells from bone marrow and adipose tissue. *J Cell Biochem* 2011;112:1206-1218.
36. Roura S, Pujal JM, Gálvez-Montón C, Bayes-Genis A. The role and potential of umbilical cord blood in an era of new therapies: A review. *Stem Cell Res Ther* 2015;6:123.
37. Park YB, Song M, Lee CH, Kim JA, Ha CW. Cartilage repair by human umbilical cord blood-derived mesenchymal stem cells with different hydrogels in a rat model. *J Orthop Res* 2015;33:1580-1586.